

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 2 – Technical Presentation

|                  |                                                                                                                                                          |               |                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------|
| Course Code:     | DSA0308                                                                                                                                                  | Course Title: | Natural Language Processing                |
| CO Assessed:     | CO1 – Imitation (L1)                                                                                                                                     | Assessment:   | Assessment Tool 2 – Technical Presentation |
| Weightage:       | 30% of CIA                                                                                                                                               |               |                                            |
| CO1 Statement:   | Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3) |               |                                            |
| Student Name:    | R.Kalyan Kumar Reddy                                                                                                                                     | Reg. No.:     | 192472274                                  |
| Section / Batch: | 2024                                                                                                                                                     | Date:         | 16/07/2026                                 |

### Code of Conduct

*I R.Kalyan Kumar Reddy - 192472274 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: kalyan

### Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

| Section / Topic                                                                 | Focus                                           | Q Nos |
|---------------------------------------------------------------------------------|-------------------------------------------------|-------|
| Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP | NLP Models, Language Processing Evolution       | TP-01 |
| Comparative Study of NLP Models and Algorithms                                  | Model Analysis, Algorithm Comparison            | TP-02 |
| Regular Expressions for Real-World Text Processing                              | Pattern Matching, Information Extraction        | TP-03 |
| Designing Efficient Regular Expression Patterns for Information Extraction      | Text Mining, Data Extraction Techniques         | TP-04 |
| Finite State Automata in Natural Language Processing                            | State Transitions, Language Recognition         | TP-05 |
| Building a Finite State Automaton for Token Recognition                         | Lexical Analysis, Token Validation              | TP-06 |
| Morphology and Word Formation in Natural Languages                              | Morpheme Analysis, Word Structure Processing    | TP-07 |
| Inflectional Morphology: Understanding Grammatical Variations                   | Grammatical Analysis, Word Normalization        | TP-08 |
| Derivational Morphology and Vocabulary Expansion                                | Word Generation, Semantic Transformation        | TP-09 |
| Comparative Analysis of Inflectional and Derivational Morphology                | Morphological Classification, Language Analysis | TP-10 |

|                                                                               |                                                |       |
|-------------------------------------------------------------------------------|------------------------------------------------|-------|
| Finite-State Morphological Parsing Techniques                                 | Morphological Parsing, Finite-State Processing | TP-11 |
| Morphological Analysis for Indian Languages                                   | Language-Specific Morphology, NLP Challenges   | TP-12 |
| Porter Stemmer: Design, Working Mechanism, and Applications                   | Stemming Algorithms, Text Preprocessing        | TP-13 |
| Stemming vs Morphological Parsing: A Comparative Study                        | Word Reduction, Morphological Processing       | TP-14 |
| Integrated NLP Pipeline: From Regular Expressions to Morphological Processing | End-to-End Language Processing Pipeline        | TP-15 |

## Annexure A – Technical Presentation

| Team No. | Technical Presentation Title                                                    | Description                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP | Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.              |
| 2        | Comparative Study of NLP Models and Algorithms                                  | Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.                             |
| 3        | Regular Expressions for Real-World Text Processing                              | Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.                 |
| 4        | Designing Efficient Regular Expression Patterns for Information Extraction      | Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.                    |
| 5        | Finite State Automata in Natural Language Processing                            | Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.                              |
| 6        | Building a Finite State Automaton for Token Recognition                         | Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.                                    |
| 7        | Morphology and Word Formation in Natural Languages                              | Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.                                             |
| 8        | Inflectional Morphology: Understanding Grammatical Variations                   | Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.                                               |
| 9        | Derivational Morphology and Vocabulary Expansion                                | Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.                                    |
| 10       | Comparative Analysis of Inflectional and Derivational Morphology                | Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.                                                   |
| 11       | Finite-State Morphological Parsing Techniques                                   | Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.                                   |
| 12       | Morphological Analysis for Indian Languages                                     | Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.        |
| 13       | Porter Stemmer: Design, Working Mechanism, and Applications                     | Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.                                      |
| 14       | Stemming vs Morphological Parsing: A Comparative Study                          | Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.                                         |
| 15       | Integrated NLP Pipeline: From Regular Expressions to Morphological Processing   | Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing. |

## Rubrics for Calculation

| Technical Presentation     |           |                                                                    |                                                                |                                                            |                                                                     |
|----------------------------|-----------|--------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Criteria                   | Weightage | Level 4 – Exemplary                                                | Level 3 – Proficient                                           | Level 2 – Developing                                       | Level 1 – Beginning                                                 |
| Technical Content Accuracy | 30        | Highly accurate, in-depth, and insightful technical explanation    | Technically correct content with adequate depth and clarity    | Basic concepts presented with limited accuracy and depth   | Content contains major technical errors; concepts poorly understood |
| Problem Identification     | 25        | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach    | Problem identified but analysis or methodology is weak     | Problem statement unclear or incorrect; no logical approach         |
| Use of Tools               | 20        | Advanced tools, well-analyzed data, and highly effective visuals   | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation         | Minimal or incorrect use of tools and data; poor visuals            |
| Communication              | 15        | Highly engaging, confident, and audience-focused delivery          | Clear, organized, and professional communication               | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides               |
| Professionalism            | 10        | Exemplary professionalism, leadership, and ethical awareness       | Professional conduct with effective team collaboration         | Basic professionalism; uneven team participation           | Lacks professionalism; minimal contribution or ethical awareness    |

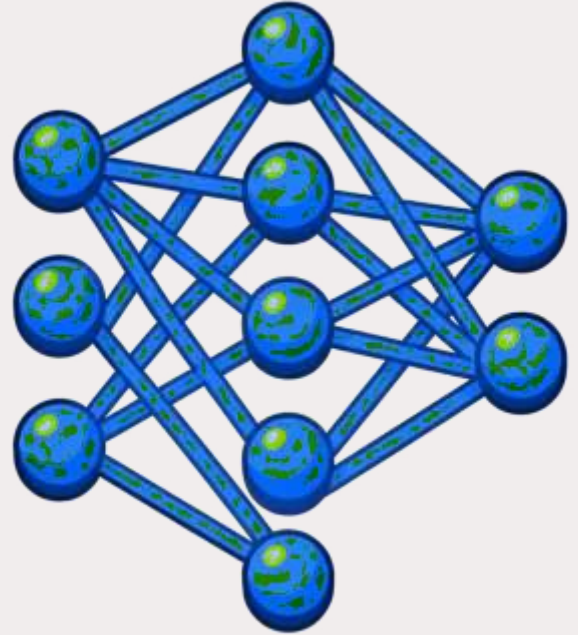
| Team No. / Criteria Level | Technical Content Accuracy | Problem Identification | Use of Tools | Communication | Professionalism | Total % |
|---------------------------|----------------------------|------------------------|--------------|---------------|-----------------|---------|
|                           | 30                         | 25                     | 20           | 12            | 7               | 94      |

| Faculty In-charge    | Course Coordinator | Program Director |
|----------------------|--------------------|------------------|
| Dr.T.Kumaragurubaran |                    |                  |
| Signature: _____     | Signature: _____   | Signature: _____ |
| Date: _____          | Date: _____        | Date: _____      |

# Language Models in Natural Language Processing (NLP)

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# What is a Language Model?



## Predicting Word Sequences

A system that learns the probabilities of word sequences — predicting what comes next in a sentence.



## Everyday Analogy

Your phone's keyboard suggests the next word as you type — that's a language model in action.



## Trained on Massive Data

Captures grammar, facts, and context by learning from enormous collections of text.

# Why Do We Need Language Models?

## Ambiguous by Nature

Human language is ambiguous, contextual, and constantly evolving — impossible to hard-code.

## Classic Example

"I saw the man with the telescope" — who has the telescope? Context decides.

## Handling Scale

Language models help machines resolve ambiguity across billions of daily interactions: search, translation, voice assistants.



# How Language Models Work



Each piece of text is broken into tokens, converted to numerical vectors, then fed through a model that predicts the next token — repeating millions of times during training to minimize errors.

# Types of Language Models

| LQpg        | KgQ Idga                  | Exemplg        |
|-------------|---------------------------|----------------|
| N-gram      | Count word co-occurrences | Spell checkers |
| RNN / LSTM  | Sequential memory         | Early chatbots |
| Transformer | Self-attention mechanism  | GPT-4, BERT    |

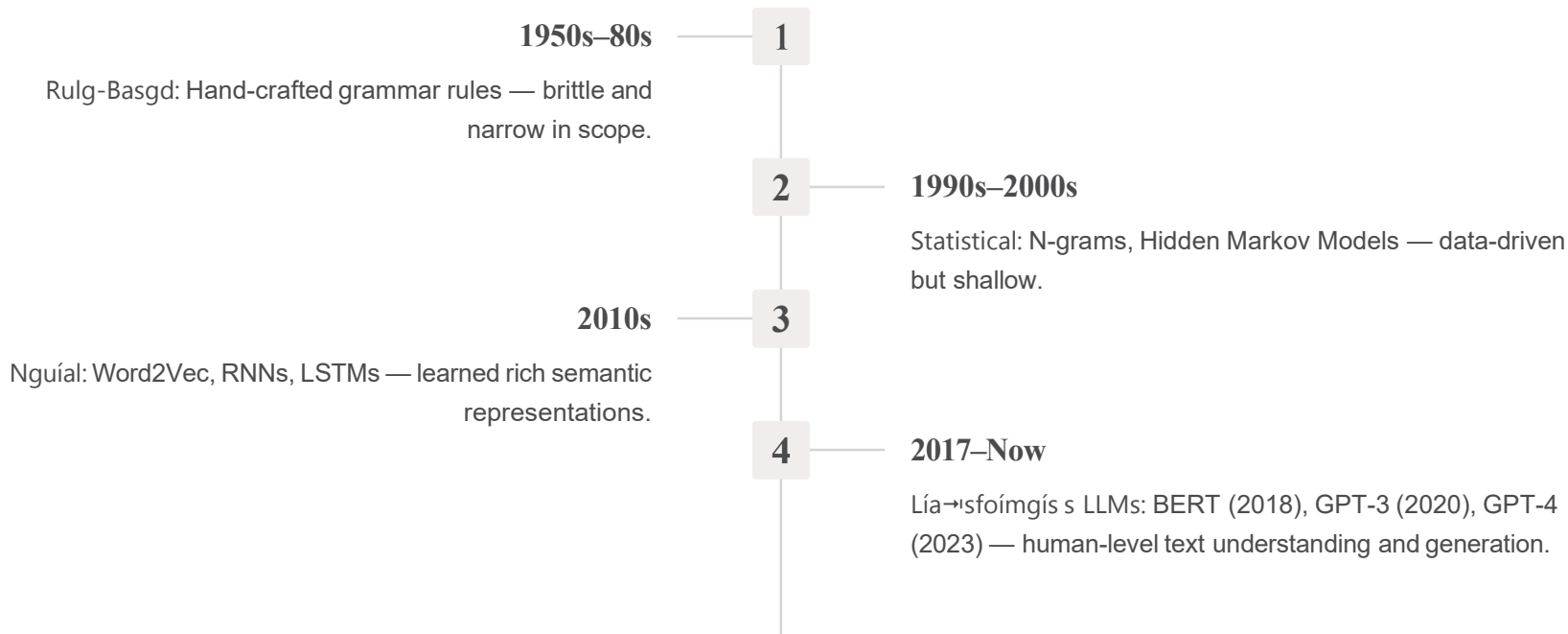
## From Simple Counts to Deep Attention

N-gram models predict the next word from just the previous N words — fast but forgetful.

RNNs and LSLMs introduced sequential memory, allowing context to persist over longer text passages.

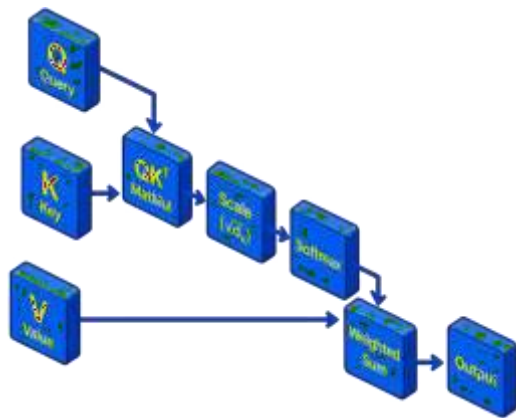
Large language models revolutionized the field by attending to all words simultaneously — today's gold standard powering GPT-4 and BERT.

# Evolution of Language Models



Language modeling has progressed from hand-crafted statistical models to large language models capable of processing, understanding, and generating human-level text at remarkable scale.

# The Transformer: The Game-Changer



Introduced in "Attention Is All You Need" (Vaswani et al., 2017)

## Self-Attention

Every word looks at every other word simultaneously to understand context.

## Resolves Ambiguity

"Bank by the river"  
vs. "Money in the bank" — attention knows the difference.

## Parallel Training

Enables training on billions of parameters in parallel — the foundation of GPT & BERT.

# Real-World Applications



## Q Search Engines

Google uses BERT to understand the true intent behind your queries.



## 🌐 Machine Translation

Google Translate handles 100+ languages in real time for billions of users.



## 💬 Chatbots h Assistants

ChatGPT, Siri, Alexa, and customer support bots all rely on language models.



## 💊 Healthcare

Extracting insights from clinical notes and medical literature at scale.



# Advantages h Challenges

## ☑ Strengths

- Fluent, coherent text generation
- Multilingual capabilities
- Few-shot learning across tasks
- Broad task coverage without retraining

## ⚠ Hallucinations

Models confidently generate false information — a critical reliability concern.

## ⚠ Bias

Training data reflects societal biases — outputs can be unfair or harmful.

## ⚠ Cost h Scale

GPT-3 training cost ~\$4.6M; acquiring general-purpose language ability requires training billions of parameters on massive text data, as predicted by scaling laws.

# The Future of Language Models



## Multimodal AI

Models that understand text, images, audio, and video together — GPT-4o leads the way.



## Smaller, Efficient Models

Running powerful LLMs on phones and edge devices — bringing AI to everyone, everywhere.



## Reasoning Agents

LLMs that plan, use tools, and take autonomous actions — moving beyond text generation.



## Responsible AI

Alignment, explainability, and reducing hallucinations are the top priorities for the field.

LLMs like ChatGPT have shown impressive capabilities in NLP tasks — and a systematic exploration of their full potential across the field is only just beginning.



# Key Takeaways

01

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## Core Principle

Language models learn to predict a word given the context from vast amounts of training data.

02

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## The Arc of Progress

Rule-based → Statistical → Neural → LLMs — each era unlocking new capabilities.

03

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## Already Around You

They power tools you use every day — search, translation, voice assistants, and more.

04

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## Exciting but Unfinished

An exciting future lies ahead — but challenges like bias and hallucinations must be solved responsibly.

[illegible]